# *COMSATS University Islamabad*

# *Attock Campus*


## *Department Of Computer Science*

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|---|---|
| *Course* | *Info. Security (Lab)* |
| *Instructor* | *Mam. Ambreen Gul* |
| *Lab Assignment* | *04* |

## *Student Details*

| *Registration No.* | *Name* |
|---|---|
| *FA24-BSE-032* | *Umair Ahmed* |

**Task 1: Simple RSA Implementation (Without Libraries)**

**Objective: Understand the math behind RSA**

**Tasks:**

- **Manually implement key generation using small primes**
- **Write functions for: o Encrypting a message (using public key) o Decrypting a message (using private key)**
- **Test it on your name or a short string**

## Code:

```python
# Choosing two prime numbers
p = 17
q = 19

# Calculate n and phi
n = p * q
phi = (p - 1) * (q - 1)

# Public key exponent (must be coprime with phi)
e = 7


# Finding modular inverse (private key d)
def mod_inverse(e, phi):
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d
    return None


# Private key
d = mod_inverse(e, phi)


# Encryption function
def encryption(message, e, n):
    encrypted = []

    for char in message:
        m = ord(char)        # Convert character to ASCII
        c = (m ** e) % n     # RSA encryption
        encrypted.append(c)

    return encrypted
```

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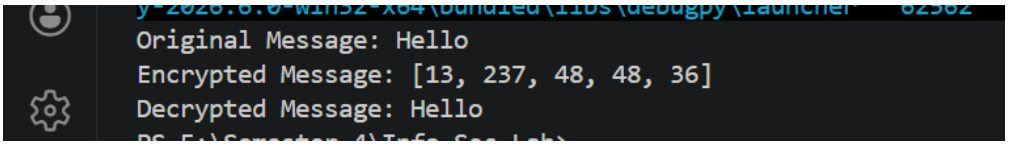
```python
def encryption(message, e, n):
        c = (m ** e) % n     # RSA encryption
        encrypted.append(c)

    return encrypted


# Decryption function
def decryption(cipher, d, n):
    decrypted = ""

    for c in cipher:
        m = (c ** d) % n     # RSA decryption
        decrypted += chr(m)  # Convert ASCII back to character

    return decrypted


# Original message
message = "Hello"

print("Original Message:", message)

# Encrypt
cipher_text = encryption(message, e, n)
print("Encrypted Message:", cipher_text)

# Decrypt
plain_text = decryption(cipher_text, d, n)
print("Decrypted Message:", plain_text)
```

## Output:



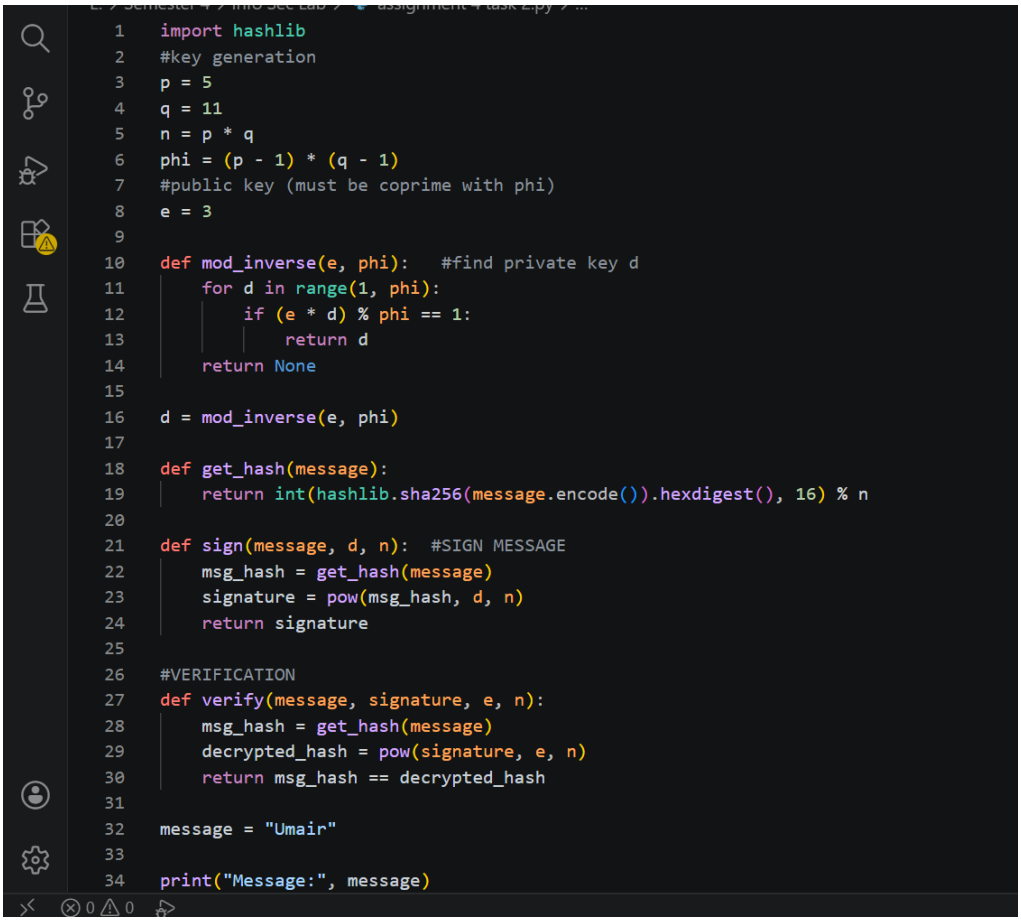

**Task 2: Create a Digital Signature**

**Objective: Understand digital signatures**

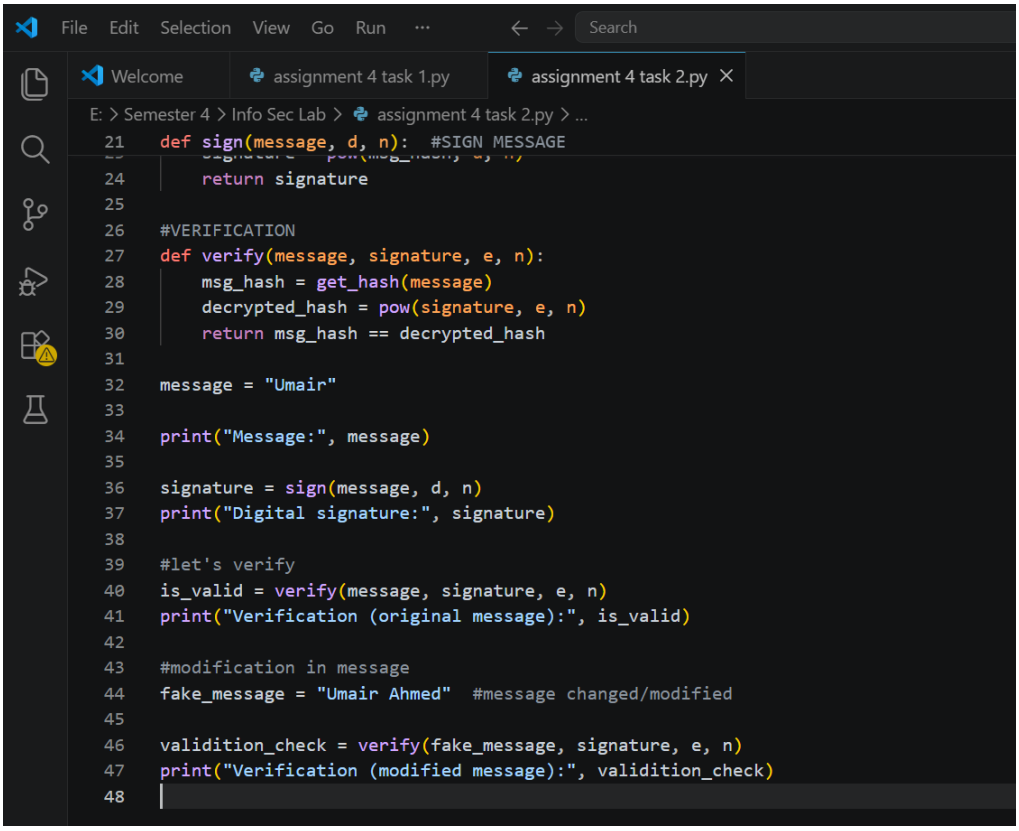
**Tasks:**

- **Generate RSA key pair**
- **Create a hash of a message (e.g., using SHA256)**
- **Sign the hash using private key**
- **Verify it using the public key**
- **Modify the message slightly and show that verification fails**

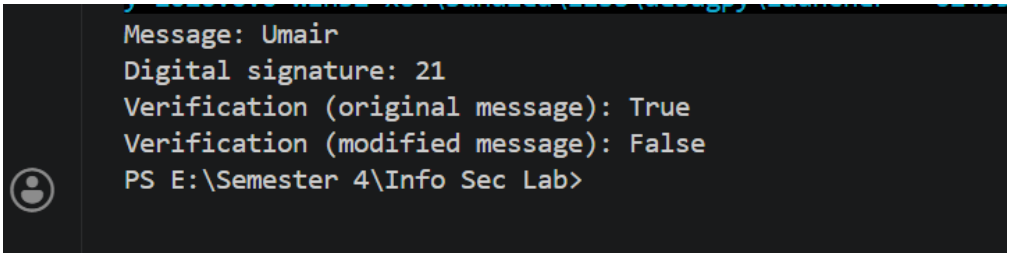
## Code:



```python
import hashlib
#key generation
p = 5
q = 11
n = p * q
phi = (p - 1) * (q - 1)
#public key (must be coprime with phi)
e = 3

def mod_inverse(e, phi):   #find private key d
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d
    return None

d = mod_inverse(e, phi)

def get_hash(message):
    return int(hashlib.sha256(message.encode()).hexdigest(), 16) % n

def sign(message, d, n):  #SIGN MESSAGE
    msg_hash = get_hash(message)
    signature = pow(msg_hash, d, n)
    return signature

#VERIFICATION
def verify(message, signature, e, n):
    msg_hash = get_hash(message)
    decrypted_hash = pow(signature, e, n)
    return msg_hash == decrypted_hash

message = "Umair"

print("Message:", message)
```


File Edit Selection View Go Run ···
Search
Welcome
assignment 4 task 1.py
assignment 4 task 2.py
E: > Semester 4 > Info Sec Lab > assignment 4 task 2.py > ...
21 def sign(message, d, n): #SIGN MESSAGE
24 return signature
25
26 #VERIFICATION
27 def verify(message, signature, e, n):
28 msg_hash = get_hash(message)
29 decrypted_hash = pow(signature, e, n)
30 return msg_hash == decrypted_hash
31
32 message = "Umair"
33
34 print("Message:", message)
35
36 signature = sign(message, d, n)
37 print("Digital signature:", signature)
38
39 #let's verify
40 is_valid = verify(message, signature, e, n)
41 print("Verification (original message):", is_valid)
42
43 #modification in message
44 fake_message = "Umair Ahmed" #message changed/modified
45
46 validition_check = verify(fake_message, signature, e, n)
47 print("Verification (modified message):", validition_check)
48

## Output:


Message: Umair
Digital signature: 21
Verification (original message): True
Verification (modified message): False
PS E:\Semester 4\Info Sec Lab>